

Fall 2025, 4100 Problem Set 2, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Given $A = \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}$ show that the set $C = \{B \in \mathbb{R}^{2 \times 2} : AB = BA\}$ is a subspaces of $\mathbb{R}^{2 \times 2}$ and find a basis.

$$\text{assume } B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 3a - c & 3b - d \\ 2c & 2d \end{bmatrix}$$

$$BA = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3a & -a + 2b \\ 3c & -c + 2d \end{bmatrix}$$

assuming $AB = BA$, we have

$$3a - c = 3a,$$

$$3b - d = -a + 2b,$$

$$2c = 3c,$$

$$2d = -c + 2d$$

$$\text{so } c = 0 \text{ and } d = a + b$$

$$\text{so } B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & a + b \end{bmatrix}$$

this shows that C is a subspace since it contains zero and is closed under addition and scalar multiplication since a and b vary freely.

it is 2 dimensional, so a convenient basis is $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right\}$

2. Let V be a vector space such that $\dim V = n$. Show that any set of n (non-zero) vectors, $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, satisfying $\mathbf{v}_i^T \mathbf{v}_j = 0$ for $i \neq j$ is a basis for V .

any set of n linearly independent vectors in an n -dimensional space is a basis

suppose $a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n = \mathbf{0}$

multiply both sides by $\mathbf{v}_k$

$$(a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n = \mathbf{0})^T \mathbf{v}_k = a_1 (\mathbf{v}_1^T \mathbf{v}_k) + \dots + a_n (\mathbf{v}_n^T \mathbf{v}_k) = 0$$

all terms are orthogonal and disappear except $a_k (\mathbf{v}_k^T \mathbf{v}_k) = 0$

since $\mathbf{v}_k$ is non zero, a_k must be zero. This holds for each $k = 1 \dots n$, so all a_k are zero

therefore all vectors are linearly independent, and since there are n linearly independent vectors in n -dimensional space, they form a basis.

3. Let $\mathbb{P}_n$ be the space of all polynomials over $\mathbb{R}$ of degree at most n . Use the Rank-Nullity theorem to show that the subset $S = \{g \in \mathbb{P}_n : f' + 4f = g \text{ for some } f \in \mathbb{P}_n\}$ is a subspace of $\mathbb{P}_n$ of dimension $n + 1$.

define function $T(f) = f' + 4f$

T is linear because $(af + bg)' + 4(af + bg) = a(f' + 4f) + b(g' + 4g)$

the set $S = \{g \in \mathbb{P}_n : g = f' + 4f \text{ for some } f \in \mathbb{P}_n\}$ is the image of T , so $S = \text{im} T$

if $f \in \ker T$ then $f' + 4f = 0$, so $\ker T = \{0\}$

by rank nullity $\dim \mathbb{P}_n = \dim \ker T + \dim \text{im} T = (0) + \dim S$

so $\dim S = \dim \mathbb{P}_n = n + 1$

thus S is a subspace of $\mathbb{P}_n$ and $\dim S = n + 1$

4. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times l}$ such that $AB\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^l$. Prove that $\text{rank}(A) + \text{rank}(B) \leq n$ using the Rank-Nullity theorem.

consider $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $B : \mathbb{R}^l \rightarrow \mathbb{R}^n$

$ABx = 0$ for all $x \in \mathbb{R}^l$ means the composition $A \circ B$ is the zero map, so every vector in image of B is sent to zero by A

$$\text{im}(B) \subseteq \ker(A)$$

$$\text{rank}(B) = \dim \text{im}(B) \leq \dim \ker(A) = \text{nullity}(A)$$

$$n = \text{rank}(A) + \text{nullity}(A) \geq \text{rank}(A) + \text{rank}(B)$$

$$\text{rank}(A) + \text{rank}(B) \leq n$$

5. Let V be a vector space with basis $\{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}\}$ and let $T \in \mathcal{L}(V, V)$ be defined by $T\mathbf{v}_i = \mathbf{v}_{n-i}$. Show that $T^{-1} = T$ and that $T - I$ is not invertible. (Here $(T - I)\mathbf{x} = T\mathbf{x} - \mathbf{x}$)

$$T^2 v_i = T(T(v_i)) = T v_{n-i} = v_{n-(n-i)} = v_i$$

since T^2 acts as the identity on every basis vector, $T^2 = I$, so $T^{-1} = T$

$$(T - I)(v_i + v_{n-i}) = T v_i + T v_{n-i} - (v_i + v_{n-i}) = v_{n-i} + v_i - (v_i + v_{n-i}) = 0$$

thus every nonzero vector of form $v_i + v_{n-i}$ lies in $\ker(T - I)$, so $\ker(T - I) \neq 0$
a linear map is invertible iff its kernel is $\{0\}$, so $T - I$ is not invertible

6. Consider the block-diagonal matrix $A = \begin{bmatrix} B & 0 \\ 0 & C \end{bmatrix}$. Show that $\text{nullity}(A) = \text{nullity}(B) + \text{nullity}(C)$.

suppose $B \in \mathbb{R}^{m \times n}$ and $C \in \mathbb{R}^{p \times q}$

then A is a block matrix of size $(m+p) \times (n+q)$

a vector in the domain of A has the form $\begin{bmatrix} x \\ y \end{bmatrix}$, $x \in \mathbb{R}^n$, $y \in \mathbb{R}^q$

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} B & 0 \\ 0 & C \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} Bx \\ Cy \end{bmatrix}$$

$$\text{so } \ker(A) = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : Bx = 0, Cy = 0 \right\} = \ker(B) \oplus \ker(C)$$

$$\dim \ker(A) = \dim \ker(B) + \dim \ker(C)$$

$$\text{nullity}(A) = \text{nullity}(B) + \text{nullity}(C)$$